

Homework Assignment 3

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1. (1.B.5) Give an example of a sequence of continuous real-valued functions $f_1, f_2, \dots$ on $[0, 1]$ and a continuous real-valued function f on $[0, 1]$ such that

$$f(x) = \lim_{k \rightarrow \infty} f_k(x)$$

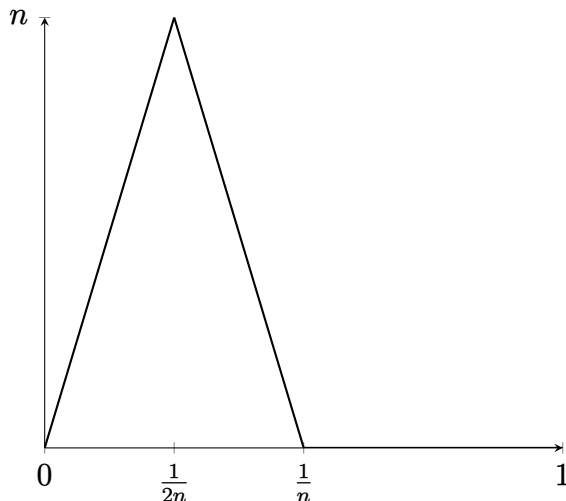
for each $x \in [0, 1]$ but

$$\int_0^1 f \neq \lim_{k \rightarrow \infty} \int_0^1 f_k.$$

Proof. Define

$$f_n(x) = \begin{cases} 2n^2x & 0 \leq x < \frac{1}{2n} \\ -2n^2x + 2n & \frac{1}{2n} \leq x < \frac{1}{n} \\ 0 & \frac{1}{n} \leq x \leq 1, \end{cases}$$

so that $\int_0^1 f_n = \frac{1}{2}$ for all $n \in \mathbb{Z}^+$ as shown below.



But the limit f_n as $n \rightarrow \infty$ is $f(x) = 0$, since $f_n(0) = f_n(1) = 0$ for all n , and for any $x \in (0, 1)$ there is an n large enough that $\frac{1}{n} < x$, so $f_m(x) = 0$ for all $m \geq n$. Since $\int_0^1 f = 0$ and $\lim_{n \rightarrow \infty} \int_0^1 f_n = \frac{1}{2}$, we have our contradiction. $\square$

2. (2.A.9) Prove that $|A| = \lim_{t \rightarrow \infty} |A \cap (-t, t)|$ for all $A \subset \mathbb{R}$.

Proof. UNFINISHED

□

3. (2.A.11) Prove that if $I_1, I_2, \dots$ is a disjoint sequence of open intervals, then

$$\left| \bigcup_{k=1}^{\infty} I_k \right| = \sum_{k=1}^{\infty} \ell(I_k).$$

Proof. NOT STARTED

□

4. (2.B.5) Suppose S is the smallest σ -algebra on $\mathbb{R}$ containing $\{(r, r+1) : r \in \mathbb{Q}\}$. Prove that S is the collection of Borel subsets of $\mathbb{R}$.

Proof. NOT STARTED

□

5. (2.B.11) Suppose $\mathcal{T}$ is a σ -algebra on a set Y and $X \in \mathcal{T}$. Let $S = \{E \in \mathcal{T} : E \subset X\}$.

(a) Show that $S = \{F \cap X : F \in \mathcal{T}\}$.

(b) Show that S is a σ -algebra on X .

Proof. NOT STARTED

□

6. (2.B.13) Suppose (X, S) is a measurable space, $E_1, \dots, E_n$ are disjoint subsets of X , and $c_1, \dots, c_n$ are distinct nonzero real numbers. Prove that $c_1\chi_{E_1} + \dots + c_n\chi_{E_n}$ is an S -measurable function if and only if $E_1, \dots, E_n \in S$.

Proof. NOT STARTED

□